# EXPERIMENT 4

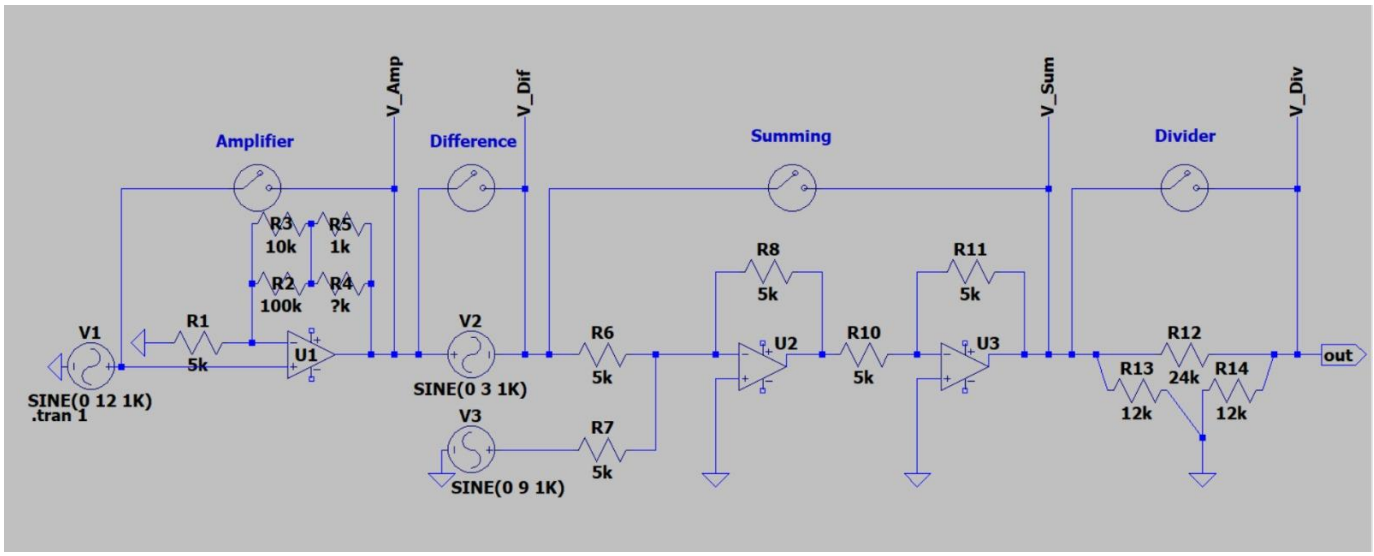
***Design of an Experiment about Fundamental Operations with Voltages***

**GROUP NO:** 1.3

*Objectives: To analyze the addition, subtraction, multiplication, and division operations of operational amplifiers and to analyze and solve circuits involving these operations.*

*Materials:*

1. 13 Resistors (6 5k, 1 10k, 2 12k, 1 24k, 1 100k,1 1k)
2. 3 Op-Amps
3. 3 Voltage Sources




*Procedure:*

1. Firstly, in the Amplifier section, there is a Wheatstone bridge structure. Using this, find the missing value of R4.
2. After finding R4 value find the current flowing through R2.
3. Afterward, in the Summing section, there is a second op-amp that inverts the negative output of the first one. Investigate ways to simplify the circuit while still obtaining a positive result and draw the final form of the circuit.
4. Apply delta-wye transformation to the divider part, then apply the changes to the circuit.

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**GROUP NO:** 1.3

Group Member 1: Erkan TİRYAKİOĞLU, 150720051, Report Preparation, Procedure.



Group Member 2: Burakhan SİNAN, 150721001, Circuit Design, Procedure.



Group Member 3: Mehmet Mete EKER, 150722013, Circuit Design, Report Preparation.



Group Member 4: Şeref KESER, 150722019, Circuit Design, Procedure.



Group Member 5: Kaan YALÇIN, 150719068, Report Preparation, Procedure.

